US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12384596
Kind Code B2
Date of Patent August 12, 2025
Inventor(s) Ogawa; Tetsuro et al.

Container and container assembly

Abstract

A container and container assembly can suppress container deformation, without use of liquid feeder container supports, even when internal container pressure becomes high during pressure feed of a liquid using high-pressure gas, whereby performing stably liquid feed. The container and container assembly have high versatility and can be used in a liquid feeder having no container support. A container has an approximately cylindrical main body portion wherein the main body portion includes a cylindrical mouth opening at one end, a stepped portion extending from the other end. The container has a reduced diameter, and round bottom portion extending from and bulging away from the stepped portion. Multiple concave portions engage with support base claw portions enabling the container to stand by itself are formed at the stepped portion and recessed at non-overlapping positions with a series of parting lines extending through the main body, stepped, and round bottom portions.

Inventors: **Ogawa; Tetsuro (Toyohashi, JP), Yamamoto; Takashi (Toyohashi, JP)**

Applicant: **AICELLO CORPORATION** (Toyohashi, JP)

Family ID: **80737852**

Assignee: **AICELLO CORPORATION (Toyohashi, JP)**

Appl. No.: **18/561003**

Filed (or PCT Filed): **March 31, 2022**

PCT No.: **PCT/JP2022/016522**

PCT Pub. No.: **WO2022/244527**

PCT Pub. Date: November 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240367857 A1	Nov. 07, 2024

Foreign Application Priority Data

JP	2021-083228	May. 17, 2021

Publication Classification

Int. Cl.: **B65D23/00** (20060101); **B65D1/02** (20060101); **B65D25/20** (20060101); **B65D25/24** (20060101)

U.S. Cl.:

CPC **B65D23/001** (20130101); **B65D1/02** (20130101); **B65D25/24** (20130101);

Field of Classification Search

CPC: B65D (1/02); B65D (1/0284); B65D (23/001); B65D (23/0878); B65D (23/12); B65D (25/24); Y10T (29/49876)

USPC: 215/395; 215/372; 215/383; 215/384; 215/376; 220/605; 220/630

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
845777	12/1906	Grossman	220/630	B65D 23/001
860316	12/1906	Nash	215/12.1	B65D 23/001
2837245	12/1957	Grebowiec	222/402.1	B65D 83/38
2905351	12/1958	Lerner	220/605	A47G 23/0216
3722725	12/1972	Khetani	220/605	B65D 23/001
3847494	12/1973	Franklin	403/345	A45D 40/0068
3927782	12/1974	Edwards	215/10	B65D 23/001
3948404	12/1975	Collins	220/605	B65D 25/24
4138026	12/1978	Conklin	215/12.2	B29C 63/423
4241839	12/1979	Alberghini	248/688	B65D 23/001
4326638	12/1981	Nickel	215/12.1	B65D 23/001
4463860	12/1983	Yoshino	220/605	B65D 23/001
4881666	12/1988	Tullman	222/548	B65D 23/02
5662241	12/1996	Sorensen	220/630	B65D 81/365
6405675	12/2001	Mills	215/DIG.7	A01K 7/00
8439223	12/2012	Smith	220/636	B65D 23/001
9061795	12/2014	Girardot	N/A	B65D 23/001
2006/0177399	12/2005	Jourdan	424/70.1	A61Q 5/06
2008/0210699	12/2007	Esteve	220/737	B65D 1/0276
2009/0184116	12/2008	Laske	220/4.28	A01K 7/00
2011/0049167	12/2010	Martin	220/630	B65D 23/001
2013/0062302	12/2012	Otero	215/12.1	B65D 11/04
2018/0215512	12/2017	Toh	N/A	N/A

2018/0296013	12/2017	Frei	N/A	A47G 23/0241
2024/0367857	12/2023	Ogawa	N/A	B65D 1/02

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102050248	12/2010	CN	N/A
107848670	12/2017	CN	N/A
211495203	12/2019	CN	N/A
2357437	12/1975	FR	B65D 81/20
S57-26337	12/1981	JP	N/A
S58-76899	12/1982	JP	N/A
H01-147983	12/1988	JP	N/A
H07-2233	12/1994	JP	N/A
2000-010390	12/1999	JP	N/A
2011-098736	12/2010	JP	N/A
2019-001508	12/2018	JP	N/A
2020-196537	12/2019	JP	N/A

OTHER PUBLICATIONS

Jun. 7, 2022 International Search Report issued in International Patent Application No. PCT/JP2022/016522. cited by applicant
Jun. 7, 2022 Written Opinion issued in International Patent Application No. PCT/JP2022/016522. cited by applicant
Mar. 26, 2025 Extended European Search Report issued in European Application No. 22804443.4. cited by applicant
Jun. 19, 2025 Office Action issued in Chinese Patent Application No. 202280036120.X. cited by applicant

---

Primary Examiner: Weinerth; Gideon R

Attorney, Agent or Firm: Oliff PLC

Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a container for containing a liquid and to a container assembly including this container.
BACKGROUND OF THE ART
(2) Liquids such as material liquids for foods and high-purity chemical solutions for industrial use (for example, photoresist and cleaning agent used in manufacture of semiconductors and manufacture of liquid crystal displays) are contained in containers while their high purities are maintained so as to prevent contamination by foreign substances or deterioration of the liquids. These liquids are stored and transported while being contained in the containers. Also, a container containing a liquid is set into a liquid feeder, and the liquid is discharged from the container and is used.
(3) In general, the bottom surface of such a container is formed to be flat so as to enable the container to stand by itself. A method for discharging a liquid by means of pressure feed is a known technique. In this method, a liquid feed pipe and a gas supply pipe are inserted into a container, and

a gas is fed into the container through the gas supply pipe so as to increase the internal pressure of the container, thereby feeding the liquid into the liquid feed pipe, whereby the liquid is discharged from the container. In this case, a high-pressure gas (for example, 100 to 200 kPa) is frequently used so as to pressure-feed a liquid which is high in viscosity and does not flow easily. In the case of a generally known container having a flat bottom surface, when the pressure inside the container increases due to supply of the high-pressure gas, the bottom surface may bulge and deform, and the container may tilt. When the amount of the liquid remaining in the container decreases, the end of the liquid feed pipe becomes unable to reach the liquid, because of the deformation of the bottom surface and the tilting of the container, possibly resulting in the inability to discharge from the container a portion of the liquid contained in the container.
(4) Patent Document 1 describes a container having a hemispherical bottom surface and a support base (cradle) for enabling the container to stand by itself. As a result of engagement between a concave groove formed on the outer surface of a circumferential wall of the container and being continuous in the circumferential direction and convex portions provided on the inner surface of a circumferential wall of the support base, the container and the support base are fitted together. Since the bottom surface of the container is hemispherical, the internal pressure of the container generated as a result of supply of the above-described high-pressure gas acts equally on the bottom surface. Since the entire bottom surface extends downward due to the pressure of the high-pressure gas, the concave groove follows this extension and extends in the manner of a helical spring. Consequently, the concave groove pushes out the convex portions of the cradle and cannot maintain the engagement with the convex portions. As a result, the container comes off the support base and falls over, and it becomes unable to stably pressure-feed the liquid.
(5) Patent Document 2 describes a liquid container which includes a container main body having a round bottom and a support base which has a penetrating portion at its center and which supports the container main body, thereby enabling the container main body to stand by itself. A circumferential groove for engagement with the support base is formed on the container main body. When this liquid container is used, it is set into a liquid feeder which includes a pedestal having a protrusion which is fitted into the penetrating portion of the support base and comes into contact with the round bottom, and a pressing member which presses an upper end of the container main body. Since the container main body is sandwiched between container supports provided in the liquid feeder, such as the protrusion of the pedestal and the pressing member, even when the internal pressure of the container main body increases, the circumferential groove does not extend and the container main body does not come off the support base. However, in the case of the liquid container described in Patent Document 2, since the liquid feeder having container supports such as the protrusion and the pressing member is indispensable to pressure-feed the liquid within the container by the high-pressure gas, limitations are imposed on liquid feeders into which the container can be set, and therefore, the liquid container described in Patent Document 2 is poor in versatility.
PATENT DOCUMENTS
(6) [Patent Document 1] JPS58-76899U [Patent Document 2] JP2011-098736A
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
(7) The present invention has been accomplished so as to solve the above-described problems, and an object of the present invention is to provide a container and a container assembly which can remarkably suppress deformation of the container, without use of container supports of a liquid feeder, even when the internal pressure of the container becomes high at the time of pressure feed of a liquid by using a high-pressure gas, whereby stable liquid feed can be performed. The container and the container assembly have a high degree of versatility such that, for their use, they can be set into a liquid feeder having no container support.
Means for Solving Problems

(8) A container of the present invention which has been accomplished so as to achieve the above-described object is a container used to contain a liquid and to pressure-feed the liquid, comprising an approximately cylindrical main body portion, a cylindrical mouth opening at one end of the main body portion, a stepped portion continuously extending from the other end of the main body portion and having a reduced diameter, and a round bottom portion continuously extending from the stepped portion and bulging away from the stepped portion, wherein a plurality of concave portions for engagement with claw portions of a support base which enables the container to stand by itself are intermittently formed at the stepped portion, the concave portions being recessed at positions which do not overlap with a series of parting lines extending through the main body portion, the stepped portion, and the round bottom portion, and wherein each of the concave portions has an approximately elliptical shape elongated in a circumferential direction of the main body portion, and the total length of the plurality of concave portions in the circumferential direction accounts for 20 to 50% of the largest peripheral length of the main body portion.
(9) In the container, the concave portions may be provided, symmetrically with respect to the parting lines, at four locations each being an angle of 20 to 45° away from the corresponding parting line or at six locations separated from one another by an angle of 60°, the angles being those about a center axis of the main body portion.
(10) In a container assembly of the present invention, the main body portion of each of the above-described containers has a trunk portion and a neck portion which is located closer to the cylindrical mouth and whose diameter is smaller than that of the trunk portion, and a handle is fitted and/or screwed onto an outer surface of the neck portion.
(11) The container assembly includes any of the above-described containers and a support base which has an opening into which the round bottom portion is fitted and claw portions which extend from a circumferential edge of the opening and are engaged with the concave portions, whereby the support base enables the container to stand by itself.
Effects of the Invention
(12) According to the container and the container assembly of the present invention, even when the internal pressure of the container becomes high at the time of pressure-feed of the liquid, the container does not extend or deform due to pressure, because the concave portions for engagement with the claw portions of the support base are intermittently formed at particular positions. Therefore, the support base does not come off the container, and therefore, pressure-feed of the liquid can be performed stably.
(13) Also, since the container itself does not extend in the direction of its center axis at the time of pressure feed of the liquid, container supports for suppressing deformation of the container are not required to be provided in liquid feeders. Therefore, the container and the container assembly can cope with various types of liquid feeders and have a high degree of versatility.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** shows a front view of a container to which the present invention is applied and an end view taken along line A-A and viewed in the arrow direction.
(2) FIG. **2** shows a perspective view and an exploded perspective view of a container assembly to which the present invention is applied.
(3) FIG. **3** is an end view showing a different example of the container to which the present invention is applied.
(4) FIG. **4** shows an end view of a container of Comparative Example 1 which falls outside the scope of the present invention.
(5) FIG. **5** shows perspective views of containers of Comparative Examples 2 and 4 which fall

outside the scope of the present invention.

MODE FOR CARRYING OUT THE INVENTION

(6) Hereunder, embodiments for practicing the present invention will be explained in detail, but the scope of the present invention is not limited to these embodiments.

(7) FIG. **1** shows one embodiment of a container **10** of the present invention. FIG. **1**(*a*) is a front view of the container **10**, and FIG. **1**(*b*) is an end view taken along line A-A and viewed in the arrow direction. The container **10** is a container for pressure feed which contains a liquid and which can endure an internal pressure of up to 200 kPa. The container **10** includes a tubular main body portion **12**, a cylindrical mouth **11** which is open at one end (upper end) of the main body portion **12**, a stepped portion **13** which is continuous from the other end (lower end) of the main body portion **12**, and a round bottom portion **15** which continuously extends away from the stepped portion **13** and bulges downward to have an approximately hemispherical shape. The internal space of the container **10** communicates with the outside through the cylindrical mouth **11**. This opening and a lower end of an inner wall of the round bottom portion **15** face each other.

(8) The main body portion **12** has an upper end portion **12***a* at which the cylindrical mouth **11** is open, a neck portion **12***b* which extends from its lower end, and a trunk portion **12***c* which extends further downward from a lower end of the neck portion **12***b*. The diameter of the trunk portion **12***c* decreases gradually toward the neck portion **12***b*. Also, external threads **12***a***1** and **12***b***1** protrude respectively from outer circumferential surfaces of the upper end portion **12***a* and the neck portion **12***b*.

(9) The container **10** is made of a resin and is formed by direct blow molding. In the direct blow molding, a tubular resin material which is formed by extruding a molten resin at high temperature and is called parison is sandwiched in split molds having engraved desired shapes to be produced. In this state, a lower end of the parison is pinched off by the molds, whereby a pinched-off portion is formed. A portion where the parison has been pinched off becomes a bottom portion of the container. Subsequently, compressed air is fed into the parison through a blow pin.

(10) As a result, the parison is bulged and pressed against the inner wall of the mold, whereby the container is formed. Parting lines PL, which are marks of the mating interface of the split molds, are formed continuously through the upper end portion **12***a*, the neck portion **12***b*, the trunk portion **12***c*, the stepped portion **13**, and the round bottom portion **15**.

(11) Since the container **10** is a product formed by direct blow molding, the stepped portion **13**, which is continuous with and directly adjacent to the round bottom portion **15**, has a slightly larger thickness in the vicinity of the parting lines PL, as compared with wall thicknesses at other locations on the same circumference. This thicker portion contributes to increasing rigidity, thereby suppressing deformation of the container.

(12) In the front view as shown in FIG. **1**(*a*), in a region from a lower end of the main body portion **12** to the stepped portion **13**, the concave portions **14** are recessed from an outer surface of the container **10** toward its center axis at four locations which are located on opposite sides of the parting lines PL and are symmetric with respect to the parting lines PL. The four concave portions **14** do not overlap with the parting lines PL and are formed at the stepped portion **13** to be arranged in a row in the circumferential direction of the container **10**. Claw portions **21** of a support base **20**, which will be described later, are engaged with the concave portions **14** (see FIG. **2**). As a result, the container **10** and the support base **20** are coupled together, whereby the container **10** stands by itself. All of the four concave portions **14** have the same shape. Each concave portion **14** has an approximately elliptical shape; i.e., its length H along the circumferential direction of the main body portion **12** (circumferential direction length H) is greater than its length along the direction of the center axis of the main body portion **12** (namely, the direction of the center axis of the container **10**) (center axis direction length V).

(13) The center point h of the circumferential direction length H of each concave portion **14** is located an angle of θ away from the position of the boundary between the main body portion **12**

and the stepped portion **13** on a corresponding parting lines PL. Namely, in a cross sectional view of the container **10** at the boundary between the main body portion **12** and the stepped portion **13**, each of the four concave portions **14** is separated and deviated, by the smallest angle θ, from a reference line Y, which passes through the center axis C of the container **10** and connects the parting lines PL. In the example shown in FIG. **1**(*b*), the angle θ is 30°.

(14) Since the plurality of concave portions **14** are not connected in series and provided intermittently as described above, even when the internal pressure of the container **10** increases due to the pressure of the high-pressure gas which is introduced at the time of pressure feed of the liquid contained in the container **10**, extension of the concave portions **14** in the direction of the center axis direction length V can be suppressed, and the engagement between the concave portions **14** and the claw portions **21** is maintained, so the support base **20** does not come off the container **10**. Also, since the concave portions **14** do not extend due to high pressure, the container **10** does not extend in the direction of the center axis (the vertical direction in FIG. **1**(*a*)), and the overall height of the container **10** hardly changes between the case where the internal pressure of the container **10** is normal pressure and the case where the internal pressure of the container **10** is high pressure. Since the container **10** can be applied to various types of liquid feeders which may or may not have container supports for suppressing deformation of containers, the container **10** has a high degree of versatility. As described above, according to the container **10** of the present invention, there do not arise a problem that, as in the case of a conventionally known container having a circumferential groove which is continuously concaved along the outer circumference of the container, the circumferential groove extends under high pressure and the support base comes off the container, and a problem that the overall height increases. In addition, the container **10** eliminates the necessity of container supports of the liquid feeder.

(15) In the case where the container **10** has four concave portions **14**, the above-described angle θ is preferably 20 to 45°, more preferably 20 to 40°, further preferably 20 to 30°, and most preferably 30°. Also, it is preferred that the plurality of angles θ are the same. When the angle θ falls within this range, the concave portions **14** can be disposed near the parting lines PL where the wall thickness is slightly larger and therefore deformation does not occur easily.

(16) In the case where, as shown in FIG. **1**(*b*), the four concave portions **14** are disposed in point symmetry with respect to the center axis C serving as a point of symmetry, in line symmetry with respect to the reference line Y serving as an axis, and/or in line symmetry with respect to the reference line X serving as an axis where the reference line X perpendicularly intersecting the reference line Y at the center axis C, since the high pressure at the time of pressure feed equally acts on each concave portion **14**, extension of the container **10** can be suppressed more effectively. In addition, when the claw portions **21** of the support base **20** provided at positions corresponding to the concave portions **14** are engaged with the concave portions **14**, the container **10** and the support base **20** can be coupled together by merely aligning a pair of the concave portion **14** and the claw portion **21** with each other.

(17) The ratio of the sum total of the circumferential direction lengths H of the concave portions **14** to a largest peripheral length D of the main body portion **12** (the longest circumferential length of the outer circumference of the main body portion **12**; see FIG. **2**(*a*)); i.e., an occupancy ratio of the circumferential direction lengths H of the concave portions **14** to the largest peripheral length D of the main body portion **12**, is represented by an expression of occupancy ratio=(H×N/D)×100, where N represents the number of the concave portions **14**. This occupancy ratio is preferably 10 to 90%, more preferably 15 to 70%, and further preferably 20 to 50%. When the occupancy ratio falls within this range, it is possible to engage the claw portions **21** with the concave portions **14** infallibly, and to suppress extension of the concave portions **14** due to application of high pressure to the interior of the container **10**. The circumferential direction length H is an outer circumferential length of the main body portion **12** which is missing due to presence of each concave portion **14**.

(18) The largest peripheral length of the main body portion **12** is determined based on the outer

diameter of the main body portion **12**, which is arbitrarily set in accordance with a required capacity of the container **10**. The capacity of the container **10** is specifically 3 to 20 L, more specifically 3 to 10 L. The outer diameter of the main body portion **12** is set, for example, to 120 to 360 mm. Notably, no particular limitation is imposed on the center axis direction length V of the concave portions **14**, and the center axis direction length V is set to 4 to 15 mm.

(19) When the liquid is to be fed by means of pressure feed, a liquid feed pipe (not shown) is inserted from the cylindrical mouth **11** straight toward the round bottom portion **15**, and its end is positioned such that a slight gap remains between the end and an inner wall surface of a top portion of the round bottom portion **15**. As the liquid is discharged from the container **10** as a result of the pressure feed, the liquid remaining in the container **10** collects at the top portion of the round bottom portion **15** having an approximately hemispherical shape. Since the end of the liquid feed pipe is positioned in the vicinity of the top portion of the round bottom portion **15**, the residual amount of the liquid can be reduced considerably.

(20) Also, even when the internal pressure of the container **10** becomes high due to the high-pressure gas which is introduced into the container **10** at the time of pressure feed, since the bottom surface of the container **10** has an approximately hemispherical shape and therefore the pressure acting on the inner wall surface of the round bottom portion **15** is dispersed, deformation of the bottom surface is suppressed. The radius of curvature of the inner wall surface of the round bottom portion **15** is preferably 1 to 5 times a value obtained by dividing the largest peripheral length D by 2π (namely, the radius of the trunk portion **12***c*), more preferably 1 to 4 times the value, and further preferably 1 to 3 times the value.

(21) FIG. **2**(*a*) shows a perspective view of a container assembly **100** of the present invention, and FIG. **2**(*b*) shows an exploded perspective view of the container assembly **100**. The container assembly **100** is composed of the container **10**, a support base **20**, and a handle **30**. Like the container **10**, the support base **20** and the handle **30** are also made of a resin.

(22) The support base **20** has a flat cylindrical shape as a whole and has an opening at its upper end. The support base **20** has an outer diameter equal to an outermost diameter of the main body portion **12** of the container **10**. Around the opening of the support base **20**, claw portions **21** are provided at positions corresponding to the concave portions **14**. The claw portions **21** extend upward from a side wall portion of the support base **20**. Each claw portion **21** is curved, in the shape of a hook, toward the center axis side of the support base **20**. Since the support base **20** is made of a resin, the claw portions **21** have a slight degree of flexibility. Therefore, when the container **10** and the support base **20** are coupled together, the claw portions **21** come into contact with an upper end portion of the round bottom portion **15** and deflect so as to slightly expand outward from the side surface of the support base **20**, and then are fitted into the recessed concave portions **14**. Since the container **10** and the support base **20** are coupled together only by engagement between the claw portions **21** and the concave portions **14**, they can be coupled and decoupled easily when necessary. Notably, the outer diameter of the support base **20** may be smaller than the outermost diameter of the main body portion **12**.

(23) The handle **30** has an annular portion **31** and an ear portion **32**. The annular portion **31** has a ring-like shape, thereby forming a circular opening. The annular portion **31** is attached to the neck portion **12***b* of the container **10** in such a manner that it surrounds the neck portion **12***b*. The ear portion **32** extends from a portion of an outer circumference of the annular portion **31** in a radial direction of the annular portion **31**. The annular portion **31** has an attachment hole **31***a* through which the neck portion **12***b* passes and an internal thread **31***b* provided on an inner wall surface of the annular portion **31**. Meanwhile, the ear portion **32** has a circular opening as a finger engagement hole **32***a* into which a finger is inserted. Planes defined by the opening circles of the attachment hole **31***a* and the finger engagement hole **32***a* are located to be orthogonal to each other. The handle **30** is detachably attached to the container **10** as a result of screw engagement between the external thread **12***b***1** and the internal thread **31***b*. The handle **30** is attached to the container **10**

when necessary. For example, when a worker carries the container assembly **100**, the worker can insert his/her finger into the finger engagement hole **32***a*, and lift and carry the container assembly **100**.
(24) The handle **30** is not molded integrally with the container **10** and is separate from the container **10**. Thus, the container **10** has a shape of point symmetry with respect to its center axis (excluding the external threads **12***a***1** and **12***b***1**). As a result, when the container **10** is produced by direct blow molding, it is possible to prevent the container **10** from having uneven thickness (excluding the bottom portion **15** and a pinch-off portion formed in its vicinity), thereby effectively suppressing deformation at the time of pressure feed.
(25) FIGS. **1** and **2** show the example in which the number of concave portions **14** is 4. However, no limitation is imposed on the number of the concave portions **14** so long as a plurality of concave portions **14** are formed. Specifically, the number of concave portions **14** may be 2 to 6. FIG. **3**(*a*) shows an example in which the number of the concave portions **14** is 2. FIG. **3**(*b*) shows an example in which the number of the concave portions **14** is 3. FIG. **3**(*c*) shows an example in which the number of the concave portions **14** is 5. FIG. **3**(*d*) shows an example in which the number of the concave portions **14** is 6. Each of these drawings is an end view which represents an end surface obtained in the same manner as in FIG. **1**(*b*). Specifically, the end surface is obtained by cutting the container **10** at a position between the main body portion **12** and the stepped portion **13**, and the obtained end surface is viewed in the direction toward the stepped portion **13**.
(26) The two concave portions **14** in FIG. **3**(*a*) are provided at positions determined such that the positions are in point symmetry with respect to the center axis C of the container **10** serving as a point of symmetry and each position is an angle θ (=45°) away from a corresponding parting line PL at the boundary between the main body portion **12** and the stepped portion **13**. The three concave portions **14** in FIG. **3**(*b*) are provided at positions which are separated from one another by an angle θ**1** (=120°) and which are symmetric with respect to the reference line X, which is perpendicular to the reference line Y. The concave portions **14** in FIG. **3**(*c*) are provided at positions which are separated from one another by an angle θ.sub.1 (=72°) and which are symmetric with respect to the reference line X. The concave portions **14** in FIG. **3**(*d*) are provided at positions which are separated from one another by an angle θ.sub.1 (=60°) and which are symmetric with respect to the reference lines X and Y. In each of the examples, the concave portions **14** are intermittently formed at equal intervals or at unequal intervals in the horizontal direction of the container **10** and do not overlap with the parting lines PL.
(27) The material of the container **10** is a thermoplastic resin. The member which constitutes the container **10** may have a single-layer structure or a multilayer structure. In both the case where the container **10** has a single-layer structure and the case where the container **10** has a multilayer structure, the thermoplastic resin which constitutes the container **10** preferably has a flexural modulus of at least 700 MPa. In the case where the container **10** has a multilayer structure, it is sufficient that resin layers which form the multilayer structure have the above-described flexural modulus as a whole. The container **10** formed of a thermoplastic resin having a flexural modulus of at least 700 MPa enables pressure-feed of the liquid by using a gas having a pressure of up to 200 kPa, without causing breakage of the container or large deformation of the container. The flexural modulus can be determined in accordance with JIS K7171 (2016).
(28) The greater the wall thickness of the container **10**, the greater the degree to which the strength of the container can be enhanced. Meanwhile, when the wall thickness is excessively large, a large amount of a raw material is needed, and the container has an excessively large weight. In view of this, it is preferred that the container **10** has a smaller wall thickness so long as the container **10** can be molded by using a thermoplastic resin having a flexural modulus of at least 700 MPa and a required strength can be secured. For example, the wall thickness of the trunk portion **12***c* of the container **10** may be 0.8 to 4 mm.
(29) The material of the container **10** may be a high-purity thermoplastic resin. In the case where

the container **10** has a multilayer structure, it is sufficient that at least the material of its inner surface is a high-purity thermoplastic resin. The container **10** formed of a high-purity thermoplastic resin is suitable for containing a liquid required to have a high degree of cleanness, such as semiconductor materials, chemical solutions for manufacture of semiconductors, and food materials. The high-purity thermoplastic resin refers to a resin which does not cause leaching of impurity particulates into the liquid contained in the container **10** in an amount in excess of a predetermined reference value. Cleanness is known as an index representing this reference value. The cleanness shows the degree to which the quality of the liquid deteriorates as a result of leaching of impurity particulates into the liquid contained in the container for a long period of time. The cleanness can be obtained by storing ultrapure water or a photoresist liquid in a test container for a certain period of time and then counting the number of particulates present in a portion (1 mL) of the liquid stored in the container. Particulates whose particle sizes are 0.3, 0.2, 0.1, and/or 0.06 μm or greater are counted in accordance with a standard to be applied. Specifically, the cleanness is defined by the following mathematical expression (1).

[Mathematicalexpression1]

(30)

$$\text{Cleanness(counts / mL)} = \frac{c\text{(counts)} \times a / 2\text{(mL)}}{b\text{(mL)} \times a\text{(mL)}} \quad (1)$$

(31) In Mathematical expression (1), "a" represents the capacity of the test container, and "b" represents the amount of liquid sampled from the test container. First, a liquid sample for measuring the initial cleanness is collected as follows. Into the test container having a capacity a (mL), ultrapure water or a photoresist liquid is poured in an amount which is half of the capacity of the test container; i.e., a/2 (mL). After shaking the test container for 15 seconds and leaving it for 24 hours, the ultrapure water or photoresist liquid is sampled as the liquid sample. A plug is attached to the container after measurement of the initial cleanness, and the container is allowed to stand for a certain period of time. Then, the container is rotated three times without generating bubbles. Subsequently, another liquid sample for measuring the cleanness after the ultrapure water or photoresist liquid having been stored is collected. The parameter "c" is a value representing the number of particulates contained in the entire sample liquid and counted by a particle counter. The initial cleanness and the cleanness after storage for the certain period of time are calculated from that numerical value in accordance with Expression (1). The smaller the numerical value of the cleanness, the smaller the degree to which the quality of the photoresist liquid has deteriorated. When the cleanness is smaller than 100 number/mL, it shows that the container has contained the photoresist liquid without deteriorating its quality. Such photoresist liquid does not lower the quality and yield of semiconductors or liquid crystal displays (LCD).
(32) A resin which is determined to satisfy the condition that a container **10** formed of that resin and used as a test container have a predetermined cleanness is chosen as a high-purity thermoplastic resin for forming the container **10**. In the case where the container **10** contains a photoresist liquid, a resin whose cleanness is smaller than 100 number/mL (one example of the predetermined reference value) is used. In other words, the high-purity thermoplastic resin is a resin which does not cause leaching of impurity particulates into the liquid in an amount in excess of the predetermined reference value. Alternatively, a resin whose cleanness is smaller than 200 number/mL may be used in accordance with a standard to be applied. Also, a resin whose cleanness is smaller than 50 number/mL, smaller than 10 number/mL, smaller than 5 number/mL, or smaller than 3 number/mL may be used. Notably, the plug (not shown) of the container is preferably formed of the high-purity thermoplastic resin.
(33) Also, instead of the cleanness, the degree of lowering of transparency of liquid (another example of the predetermined reference value) may be used to define the degree of leaching of impurity particulates.
(34) Examples of the resin used to from the container **10** include polyolefins such as polyethylene and polypropylene, polyamide, polyvinyl alcohol, poly (ethylene-co-vinyl alcohol), polyester, and

polyphenylene oxide. One or more of these resins may be used to form a single-layer container, and a plurality of these resins may be used to form a multilayer container. Among them, polyethylene is preferred. Specific examples include linear polyethylene (LLDPE) and high-density polyethylene (HDPE), which are copolymers of ethylene and α-olefin. From the viewpoint of rigidity and cleanness, it is preferable to form the container **10** with high-density polyethylene. From the viewpoint of environmental protection, it is preferable to use a resin that can be recycled as a material.

(35) The melt flow rate of high-density polyethylene is preferably 0.01 to 3.0 g/10 minutes, more preferably 0.05 to 2.0 g/10 minutes. The density of high-density polyethylene is preferably 0.940 to 0.970 g/cm.sup.3, more preferably 0.950 to 0.960 g/cm.sup.3. The melt flow rate can be determined in accordance with JIS K6760 (1995).

(36) At least the inner wall surface of the container **10** may be made of a polyethylene or ethylene α-olefin copolymer resin having a density of 0.940 to 0.970 g/cm.sup.3. This resin is preferred such that the weight average molecular weight measured by gel permeation chromatography is 10×10.sup.4 to 30×10.sup.4, the amount of polymer having a molecular weight of 1×10.sup.3 or less is less than 2.5% by mass, and the amounts of a neutralizer, an antioxidant, and a light stabilizer, which are quantified by liquid chromatography, are 0.01% by mass or less. Here, the α-olefin may be at least one selected from the group consisting of propylene, butene-1, 4-methyl-pentene-1, hexene-1, and octene-1. Use of such a resin enables production of a container **10** which exhibits high mechanical strength and is excellent in handling and in which leaching of impurity particulates into the contained liquid is considerably suppressed.

(37) The material of the container **10** may be a resin composition which includes a polyethylene or ethylene/α-olefin copolymer resin having a density of 0.940 to 0.970 g/cm.sup.3; a neutralizer, an antioxidant, and a light stabilizer; a light blocking pigment containing an inorganic pigment and/or an organic pigment; and a dispersant which is an olefin polymer having a number average molecular weight of 2×10.sup.3 or more. This resin preferably has a weight average molecular weight of 10×10.sup.4 to 30×10.sup.4 measured by gel permeation chromatography, and the amount of polymer having a molecular weight of 1×10.sup.3 or less is less than 5% by mass. The amounts of the neutralizer, antioxidant, and light stabilizer in the resin composition are preferably 0.01% by mass or less, respectively. At least one selected from titanium oxide, carbon black, and red iron oxide are used as the above-described inorganic pigment. At least one selected from phthalocyanine-type, quinacridone-type, and azo-type organic pigments are used as the above-described organic pigment. The amount of the light blocking pigment in the resin composition is preferably 0.01 to 5% by mass. The amount of the dispersant of the olefin-based polymer is preferably less than 5% by mass. Use of such a resin composition enables obtainment of a container **10** which exhibits high mechanical strength and is excellent in handling, in which leaching of impurity particulates into the contained liquid is considerably suppressed, and which can prevent deterioration of the liquid caused by light. Such container **10** is suitably used for storing a chemical solution for manufacture of semiconductors or the above-mentioned solvent for production of pharmaceuticals.

(38) The container **10** may have a layer structure having an inner layer, an intermediate layer, and an outer layer. In this case, the inner layer is preferably formed of a high-purity resin which includes at least one of an olefin polymer (e.g., ethylene, propylene, butene-1, 4-methyl-pentene-1, hexene-1, or octene-1) and a copolymer composed of ethylene and olefin other than ethylene; and a neutralizer, an antioxidant, and a light stabilizer. In this case, the amounts of the neutralizer, the antioxidant, and the light stabilizer are preferably 0.01% by mass at a maximum, respectively. The intermediate layer preferably contains a solvent barrier resin containing poly(ethylene-co-vinyl alcohol). Further, an adhesive resin layer formed of maleic acid modified polyethylene or the like may be provided between the inner layer and the intermediate layer and/or between the intermediate layer and the outer layer. The outer layer preferably includes a resin composition

containing a light-blocking substance. This resin composition may contain a pigment dispersant in an amount of less than 5% by mass and a light-blocking pigment in an amount of 0.01 to 5% by weight. The pigment dispersant is made of an olefin-based polymer such as polyethylene and polypropylene having a number average molecular weight of 2×10.sup.3 or more. The light-blocking pigment contains an inorganic pigment and/or an organic pigment. This resin composition may also contain an UV absorber in an amount of less than 2.5% by weight. According to such a layer structure, since particulates and metal ions do not leach from the container **10** during storage and transportation of the liquid, the quality of the high-purity liquid can be maintained, and the container **10** that is difficult to break and lightweight is obtained.
(39) No particular limitation is imposed on the materials of the support base **20** and the handle **30**, and the materials may be the same or different from the material of the container **10**. Examples of the materials include a homopolymer and/or a copolymer and/or a polymer blend, each of which includes at least one of the group consisting of low-density polyethylene, medium-density polyethylene, high-density polyethylene, linear low-density polyethylene, polypropylene, polybutene, polystyrene, polyvinyl acetate, poly(methyl methacrylate), poly(ethyl methacrylate), poly(acrylic acid), cyclic polyolefin, polyacrylonitrile, polyamide (nylon), polyesters such as polyethylene terephthalate and polybutylene terephthalate, polyurethane, polycarbonate, polyimide, polyphenylene sulfide, and poly(vinyl chloride).
(40) As a method for manufacturing the container **10**, direct blow molding (extrusion blow molding) is mentioned, but instead of this, known blow molding such as injection blow molding, multilayer extrusion blow molding, and stretch blow molding may be employed. The method for manufacturing the support base **20** and the handle **30** is preferably injection molding.
(41) Examples of the liquid contained in the container **10** include liquid chemicals such as methanol, ethanol, isopropanol, isobutanol, ethylene glycol, acetone, ethyl acetate, toluene, dimethylformamide, ethylene glycol acetate, methoxypropyl acetate, and butyl cellosolve; chemical solutions for manufacture of semiconductors and liquid crystal devices such as photoresist liquids and cleaning agents; medical and pharmaceutical solutions such as disinfectants, infusions, dialysates, and formulation raw materials; and chemical solutions for food industry such as flavorings, concentrates, and food additives.
DESCRIPTION OF EMBODIMENT
(42) The present invention will now be described specifically by referring to examples. However, the present invention is not limited to these examples.
Example 1
(43) High-density polyethylene resin (melt flow rate: 0.3 g/10 min, density: 0.951 g/cm.sup.3, flexural modulus: 1370 MPa), serving as the material of the container **10**, was hot-melted and extruded by an extruder to form a parison. The parison was sandwiched in a two-part mold and direct blow molding was performed, thereby fabricating a container **10** of Example 1. The container **10** had four concave portions **14**. Each of the four concave portions **14** had a circumferential direction length H of 25 mm and a center axis direction length V of 6 mm and was 30° away from the position of the boundary between the main body portion **12** and the stepped portion **13** on the corresponding parting line PL. The largest peripheral length D of the main body portion **12** was is 160π mm, and the proportion of the circumferential direction lengths H of the four concave portions **14** to the largest peripheral length D, i.e. the occupancy ratio (H×N/D×100), was 20%.
(44) The support base **20** and the handle **30** were fabricated by injection molding using linear low-density polyethylene resin (melt flow rate: 4.0 g/10 min, density: 0.938 g/cm.sup.3). The support base **20** was coupled to the container **10**, and the handle **30** was attached to the container **10** by screwing, thereby fabricating the container assembly **100**.
Measurement of Shape Change at High Pressure
(45) The length of the container **10** from the upper end of the main body portion **12** to the lower

end of the round bottom portion **15** was measured, as the overall height of the container **10** at normal pressure, by using a height gauge (manufactured by Mitutoyo Co., Ltd.). A pressurization device was attached to the cylindrical mouth **11** and the upper end portion **12***a*, and the internal pressure of the container **10** was increased to 200 kPa, which was kept constant for 1 hour. After 1 hour, the overall height of the container **10** was measured again and recorded as the overall height at high pressure. The ratio of the overall height at high pressure to the overall height of container **10** at normal pressure was obtained as an overall height change ratio. The obtained overall height change ratio was 0.61%.

Measurement of Amount of Residual Liquid

(46) Tap water (2 L) was stored in the container **10**, and a liquid feed pipe (not shown) was inserted straight from the cylindrical mouth **11** into the container **10** and was fixed such that its end was located near the inner wall surface of the round bottom portion **15**. Tap water was discharged from the container **10** by applying a pressure of 0.05 MPa to the interior of the container **10**. The application of the pressure was stopped when tap water no longer came out of the liquid feed pipe, and the amount of tap water remaining in the container **10** was measured. The measured amount was 0.3 mL.

Example 2

(47) A container **10** of Example 2 was fabricated by performing the same process as in Example 1 except that the circumferential direction length H of each concave portion **14** was changed to 35 mm and the occupancy ratio of the four concave portions **14** was changed to 28%. Also, the support base **20** and the handle **30** were fabricated in the same manner as in Example 1. The support base **20** was coupled to the container **10**, and the handle **30** was attached to the container **10** by screwing, thereby fabricating the container assembly **100**. For the container **10**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 0.69%.

Example 3

(48) A container **10** of Example 3 was fabricated by performing the same process as in Example 1 except that six concave portions **14** was formed at 60° intervals such that the concave portions **14** were separated from one another by 60° and their occupancy ratio was changed to 30%. Also, the support base **20** and the handle **30** were fabricated in the same manner as in Example 1. The support base **20** was coupled to the container **10**, and the handle **30** was attached to the container **10** by screwing, thereby fabricating the container assembly **100**. For the container **10**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 0.69%.

Example 4

(49) A container **10** of Example 4 was fabricated by performing the same process as in Example 3 except that the circumferential direction length H of each concave portion **14** was changed to 35 mm and the occupancy ratio of the six concave portions **14** was changed to 42%. Also, the support base **20** and the handle **30** were fabricated in the same manner as in Example 1. The support base **20** was coupled to the container **10**, and the handle **30** was attached to the container **10** by screwing, thereby fabricating the container assembly **100**. For the container **10**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 0.86%.

Example 5

(50) A container **10** of Example 5 was fabricated by performing the same process as in Example 3 except that the largest peripheral length D was change to 360 πmm, the circumferential direction length H of each concave portion **14** was changed to 50 mm, thereby changing the occupancy ratio of the six concave portions **14** to 27%, and the center axis direction length V was changed to 10 mm. Also, the support base **20** and the handle **30** were fabricated in the same manner as in Example 1. The support base **20** was coupled to the container **10**, and the handle **30** was attached to the

container **10** by screwing, thereby fabricating the container assembly **100**. For the container **10**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 1.03%. Also, the amount of residual liquid measured by performing the same operation as Example 1 was 0.6 mL.

Comparative Example 1

(51) A container **10** of Comparative Example 1 was fabricated by performing the same process as in Example 1 except that, as shown in FIG. **4**, a singe concave portion **14** was disposed to be located an angle θ (=90°) away from the position of the boundary between the main body portion **12** and the stepped portion **13** on a parting line PL, and the occupancy ratio of the single concave portion **14** was changed to 5%. Also, the support base **20** and the handle **30** were fabricated in the same manner as in Example 1. The support base **20** was coupled to the container **10**, and the handle **30** was attached to the container **10** by screwing, thereby fabricating the container assembly **100**. When the internal pressure of the container **10** was increased to 200 kPa by performing the same operation as in Example 1, the support base **20** came off the container **10**. Therefore, the overall height change ratio was not measured.

Comparative Example 2

(52) As shown in FIG. **5**(*a*), a container **40** of Comparative Example 2 was fabricated by performing the same process as in Example 1 except that, instead of the concave portions **14**, a circumferential groove **44** is provided over the entire circumference of a lower end of the main body portion **42**, thereby changing the occupancy ratio of the circumferential groove **44** to 100%, and the container **40** and a handle **40***a* were molded integrally. A support base **20** fabricated by performing the same process as in Example 1 was coupled to the container **40**, thereby fabricating the container assembly **100**. For the container **40**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 5.11%. Also, the amount of residual liquid measured by performing the same operation as Example 1 was 2.2 mL.

Comparative Example 3

(53) A container **40** of Comparative Example 3 was fabricated by performing the same process as in Comparative Example 2 except that the largest peripheral length D was change to 360π mm. A support base **20** fabricated by performing the same process as in Comparative Example 2 was coupled to the container **40**, thereby fabricating the container assembly **100**. For the container **40**, a shape change at high pressure was measured by preforming the same operation as in Example 1 and the overall height change ratio was obtained. The obtained overall height change ratio was 7.37%.

Comparative Example 4

(54) A container **40** of Comparative Example 4 capable of standing by itself without use of a support base was fabricated by performing the same process as in Comparative Example 2 except that, as shown in FIG. **5**(*b*), the concave portions **14** were not provided and a flat bottom portion **45** was formed instead of the round bottom portion **15**. When the internal pressure of the container **40** was increased to 200 kPa by performing the same operation as in Example 1, the flat bottom portion **45** bulged and the container **40** fell over. The overall height change ratio of the fallen-over container **40** was obtained. The obtained overall height change ratio was 4.10%.

(55) Table 1 collectively shows the configurations of the containers **10** of Examples 1 to 5 and the containers of Comparative Examples 1 to 4 and also shows the results of measurement of their shape changes at high pressure and measurement of their residual liquid amounts.

(56) TABLE-US-00001 TABLE 1 Examples Comparative Examples 1 2 3 4 5 1 2 3 4 Largest peripheral 160π 160π 160π 160π 360π 160π 160π 360π 160π length D (mm) Concave Number 4 4 6 6 6 1 Entire Entire 0 portions circum- circum- ference ference Position Each 30° Each 30° 60° 60° 60° 90° — — — from PL from PL intervals intervals intervals from PL

Circumferential 25 35 25 35 50 25 — — — — direction length H (mm) Center axis 6 6 6 6 10 6 6 6 — direction length V (mm) Occupancy 20 28 30 42 27 5 100 100 — ratio of concave portions (%) Handle Attached Attached Attached Attached Attached Attached Integral Integral Integral by by by by by by molding molding molding screwing screwing screwing screwing screwing screwing Overall height change 0.6 0.69 0.69 0.86 1.03 Not measured 5.11 7.37 4.10 ratio (%) because of coming off of support base Residual liquid 0.3 — — — 0.6 — 2.2 — — amount (mL)

(57) As can be understood from Table 1, in each of the containers **10** of Examples 1 to 5, the concave portions **14** did not extend. Therefore, even when the internal pressure of each container **10** was high, its overall height hardly changed although container supports for suppressing a change in the overall height of the container **10** were not used. Meanwhile, in the case of Comparative Example 1, since the container **10** had only one concave portion **14**, the engagement between the container **10** and the support base **20** was instable, and therefore, the container **10** came off the support base **20**. In the case of Comparative Examples 2 and 3 in which, instead of the concave portions **14**, the circumferential groove **44** was continuously provided over the entire circumference of the container **40**, the overall height of the container increased considerably, because the circumferential groove **44** extended in the manner of a helical spring as a result of an increase in the internal pressure of the container **40**.

INDUSTRIAL APPLICABILITY

(58) The container and the container assembly of the present invention are suitably used for pressure feed of liquids, for example, in the field of foods, in the field of pharmaceuticals, and in the field of manufacture of semiconductors and liquid crystal devices, in which clean liquids are demanded.

EXPLANATIONS OF LETTERS OR NUMERALS

(59) **10**: container, **11**: cylindrical mouth, **12**: main body portion, **12***a*: upper end portion, **12***a***1**: external thread, **12***b*: neck portion, **12***b***1**: external thread, **12***c*: trunk portion, **13**: stepped portion, **14**: concave portion, **15**: round bottom portion, **20**: support base, **21**: claw portion, **30**: handle, **31**: annular portion, **31***a*: attachment hole, **31***b*: internal thread, **32**: ear portion. **32***a*: finger engagement hole. **40**: container. **40***a*: handle. **42**: main body portion. **44**: circumferential groove. **45**: flat bottom portion. **100**: container assembly. C: center axis. D: largest peripheral length. H: circumferential direction length, h: center point. PL: parting line. V: center axis direction length. X and Y: reference lines, and θ and θ.sub.1: angles

Claims

1. A container used to contain a liquid and to pressure-feed the liquid, comprising an approximately cylindrical main body portion, a cylindrical mouth opening at one end of the main body portion, a stepped portion continuously extending from the other end of the main body portion and having a reduced diameter, and a round bottom portion continuously extending from the stepped portion and bulging away from the stepped portion, the container wherein a plurality of concave portions for engagement with claw portions of a support base which enables the container to stand by itself are intermittently formed at the stepped portion, the concave portions being recessed at positions which do not overlap with a series of parting lines extending through the main body portion, the stepped portion, and the round bottom portion; and each of the concave portions has an approximately elliptical shape elongated in a circumferential direction of the main body portion, and the total length of the plurality of concave portions in the circumferential direction accounts for 20 to 50% of a largest peripheral length of the main body portion.

2. The container according to claim 1, wherein the concave portions are provided, symmetrically with respect to the parting lines, at four locations each being an angle of 20 to 45° away from the corresponding parting line or at six locations separated from one another by an angle of 60°, the

angles being those about a center axis of the main body portion.

3. A container assembly wherein the main body portion of the container according to claim 1 has a trunk portion and a neck portion which is located closer to the cylindrical mouth and whose diameter is smaller than that of the trunk portion, and a handle is fitted and/or screwed onto an outer surface of the neck portion.

4. A container assembly comprising the container according to claim 1 and a support base which has an opening into which the round bottom portion is fitted and claw portions which extend from a circumferential edge of the opening and are engaged with the concave portions, whereby the support base enables the container to stand by itself.

5. A container assembly wherein the main body portion of the container according to claim 2 has a trunk portion and a neck portion which is located closer to the cylindrical mouth and whose diameter is smaller than that of the trunk portion, and a handle is fitted and/or screwed onto an outer surface of the neck portion.

6. A container assembly comprising the container according to claim 2 and a support base which has an opening into which the round bottom portion is fitted and claw portions which extend from a circumferential edge of the opening and are engaged with the concave portions, whereby the support base enables the container to stand by itself.